Kshitij Kanade

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SUMMARY

M.S. Electrical & Computer Engineering student at UC San Diego with experience in C/C++ software development, numerical computing, embedded systems, and technical research. Currently exploring parallel computing through CPU, OpenMP, SIMD, and CUDA implementations, with an interest in performance analysis and hardware–software co-design. Brings a cross-disciplinary foundation in software, hardware, and system-level problem-solving.

EDUCATION

University of California, San Diego — La Jolla, California, US
Master of Science, Electrical and Computer Engineering, **Track: Computer Engineering** — June 2028
Relevant Coursework: VLSI Digital System Algorithms and Architectures, Principles of Computer Architecture, Optimization and Acceleration of Deep Learning on Various Hardware Platforms, Low Power VLSI Implementation for Machine Learning.

National Institute of Technology, Tiruchirappalli — Trichy, India
Bachelor of Technology, Electrical and Electronics Engineering, **GPA: 9.13/10** — July 2025
Relevant Coursework: Digital System Design & HDLs, VLSI Design, Data Structures & Algorithms, Networks & Linear Systems.

WORK EXPERIENCE

Modelling and Control of Velocity Skin Effect and Plasmas Observed in EM Railguns, IIT Bombay | [Cert] — Mumbai, India
Project Technical Assistant under Dr. Himanshu J. Bahirat, Electrical Dept., IIT Bombay — Feb 2026 – May 2026

- Built 3+ modular Python FEM solvers and benchmarked six sparse solvers (including LU, GMRES, CG, and PARDISO), reducing memory overflow by approximately 25%
- Evaluated 15+ COMSOL configurations and 10+ meshing strategies, improving EC–MF simulation stability by ~40% and mesh quality near curved conductors by ~35%.
- Diagnosed 20+ convergence failures (ill-conditioning, NaN/Inf states) across 30+ EM field analyses to stabilize current density models.

Vajra Cybersecurity Lab, A91 Eco Hub, TCAAI, IIT Bombay | [Cert] — Mumbai, India
Senior Research Intern under Dr. Manjesh Kumar Hanawal, IEOR Dept., IIT Bombay — Dec 2024 – June 2025

- Developed cross-platform EDR features; converted 1,000+ C++ rules into optimized SQL queries, boosting detection efficiency by 9.5% and cutting backend alert runtime from 30 min to <7 s.
- Automated MITRE Caldera validation workflows using Python and Selenium, reducing execution time from 10 min to <3 min.

Centre for Excellence in Robotics, IIT Kharagpur | [Cert] — Kharagpur, India
Summer Research Intern under Dr. D. K. Pratihar, Mech. Dept., IIT Kharagpur — May 2024 – July 2024

- Integrated Intel RealSense D455 depth cameras and force sensors onto a 6-DOF robotic arm, dropping inference latency from 500 ms to 80 ms with a 1.5 N auto-stop grip.
- Upgraded Agricultural Robot using YOLOv8 and Mask R-CNN, achieving 95% navigation accuracy and improving fruit detection and picking precision by 8%.

TECHNICAL PROJECTS

Parallel Computation of AXPY | C++, CUDA, OpenMP, SIMD | [GitHub Repo] — Aug 2026 – Ongoing

- Developing and comparing serial CPU, OpenMP, SIMD-vectorized, and CUDA implementations of AXPY (`y = αx + y`).
- Setting up a Windows/MSVC and CUDA toolchain on an NVIDIA RTX 3050, with reusable infrastructure to measure execution time and evaluate memory-bandwidth performance.

LEADERSHIP EXPERIENCE

PSI Racing Club – *DAQ & E-Powertrain Head / DAQ Engineer* — Trichy/July 2023 - July 2025

- Architected distributed sensing and telemetry hardware across dual microcontrollers (ESP32 for high-speed sensor computation, Arduino Uno for display interfacing on Adafruit), establishing a reliable UART communication bridge for real-time driver telemetry.
- Designed multi-sensor signal processing pipelines on ESP32 to compute velocity, 3-axis acceleration, and fuel levels in real time, transmitting live telemetry data over Zigbee to pit-crew ground stations for performance analysis.
- Integrated CAN bus communication with BMS for eBAJA alongside HV/GLV electrical systems, motor controllers, and accumulators; implemented ML-based battery SoC and thermal prediction models (Achieved AIR 2 nationally in mBAJA Phase 1).

TECHNICAL SKILLS AND CERTIFICATIONS

Languages: C/C++, Python, Verilog/VHDL, SQL, MATLAB, Java, MicroPython, TypeScript
Hardware & Systems: COMSOL Multiphysics, ANSYS, CREO, LTSpice, OpenMP, Git, Linux/Windows Toolchains
Numerical & ML Frameworks: TensorFlow, Keras, OpenCV, PySparse, PARDISO, GMRES/CG/BiCGStab Solvers, NumPy,SciPy
Certifications: Amazon ML Summer School 2024, Machine Learning Specialization (Stanford Online, Coursera 2023), Python for Everybody (University of Michigan, Coursera 2023), Data Analytics with Python (NPTEL), Adhyayan's ACA Summer School 2023 (IIT Kanpur - Deep Learning for Computer Vision, Natural Language Processing), Introduction to Embedded Machine Learning (Edge Impulse, Coursera 2026)